

Cybersecurity Trends, Zero Trust, and Cloud-Native Security

Assoc. Professor Darwin Kim Bulawan



Course Map

Week Number	Inclusive Dates	Module Number	Assessments	Topics
1	14 Feb	1	-	Course Introduction, Discussion of Deliverables, Framework for Analyzing IT Trends, Global IT Landscape
2-3	21 Feb, 28 Feb	2	Analysis Paper (I)	Artificial Intelligence (AI) & Generative AI, Large Language Models, Enterprise AI Adoption, AI Ethics, Bias and Explainability, AI Regulation (EU AI Act, PH AI initiatives)
4-5	7 Mar, 14 Mar	3	Presentation (G)	Cybersecurity Trends, Zero Trust, and Cloud-Native Security
6-7	21 Mar, 28 Mar	4	Review Paper (I)	Data Governance, Data Privacy (Philippine DPA updates), Data Ethics & Responsible Analytics, Blockchain & Web3, Smart Contracts, Enterprise Adoption vs Speculation
8	4 Apr	5	Plan Report (G)	Internet of Things (IoT), Smart Systems & Industry 4.0, IoT Security Challenges
9	11 Apr	-	Exam	MIDTERM EXAMINATION
10-11	18 Apr 25 Apr	6	Research Paper (I)	Automation & Hyperautomation, RPA & Process Mining, Low-Code / No-Code Platforms, Quantum Computing, Post-Quantum Cryptography
12-13	2 May 9 May	7	Proposal Paper (G)	Sustainable IT, Green Computing, ESG & Digital Responsibility, DevOps & DevSecOps, Platform Engineering, GitOps
14-15	16 May 23 May	8	Analysis Report (I)	Extended Reality (AR/VR/MR), Spatial Computing, Human-Computer Interaction Trends, Digital Transformation in the Philippines, eGovernment, FinTech & Digital Banking
16-17	30 May 6 Jun	9	Presentation (G)	IT Project Failures, Governance Breakdowns, Ethical and Strategic Risks
18	13 Jun	-	Exam	FINALEXAMINATION



Learning Objectives

In the end of this session, students will be able to:

- Explain the modern cybersecurity threat landscape
- Identify common cyberattack techniques
- Describe the Zero Trust security model
- Understand cloud security challenges
- Analyze enterprise cybersecurity strategies



The Modern Cyber Threat Landscape

The internet has expanded attack opportunities dramatically. Attackers today operate like organized businesses.

Cyber Threat Landscape:

- Global network of cybercriminal groups
- Financially motivated attacks dominate
- Increasing automation of hacking tools
- Targets include governments and businesses

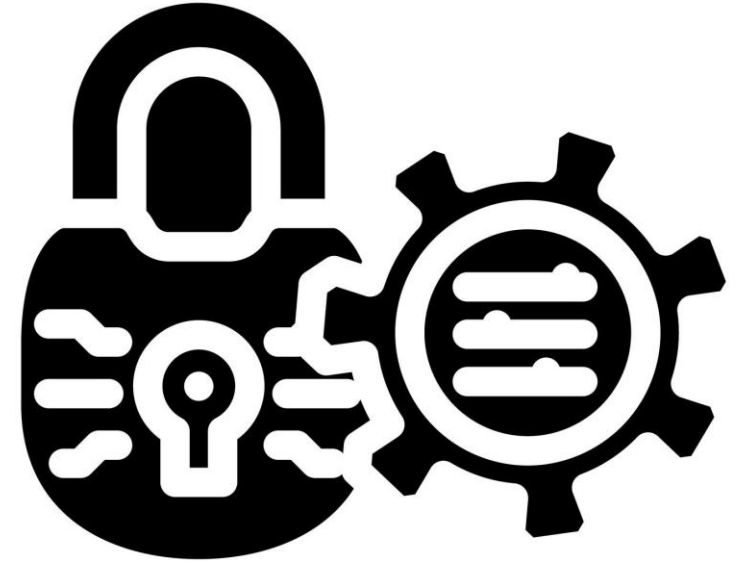


What is Cybersecurity?

Cybersecurity protects digital systems from unauthorized access and damage.

Cybersecurity:

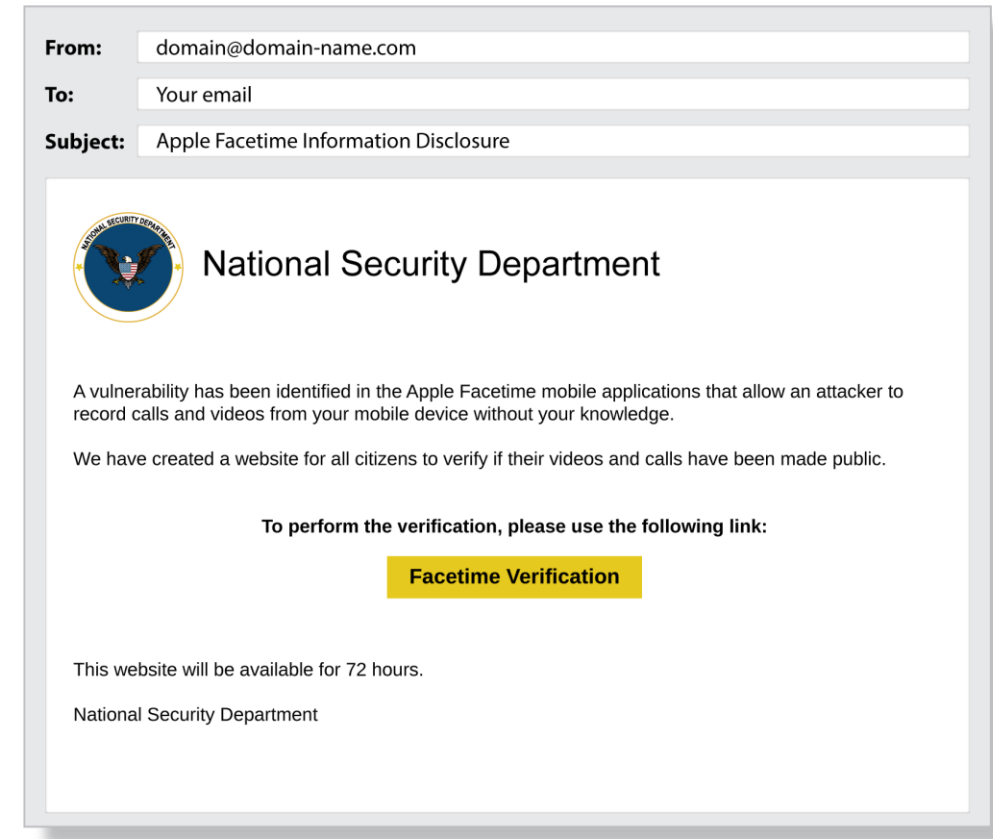
- Protection of networks, systems, and data
- Prevents unauthorized access
- Ensures confidentiality, integrity, availability



Common Types of Cyber Attacks

Cyber attackers use multiple strategies to compromise systems.

- **Phishing** – fake messages to steal information
- **Malware** – malicious software infection
- **Ransomware** – encrypting data for payment
- **Denial of Service** – overwhelming systems



Ransomware Attacks

Ransomware is one of the fastest growing cyber threats.

Ransomware:

- Malware that encrypts files
- Attackers demand payment for decryption
- Often targets hospitals, companies, governments



Human Error in Cybersecurity

Many cyber incidents occur due to simple human mistakes.

Human Security Risks:

- Weak passwords
- Clicking suspicious links
- Misconfigured systems
- Lack of security awareness

Password

— — — weak



— — — strong



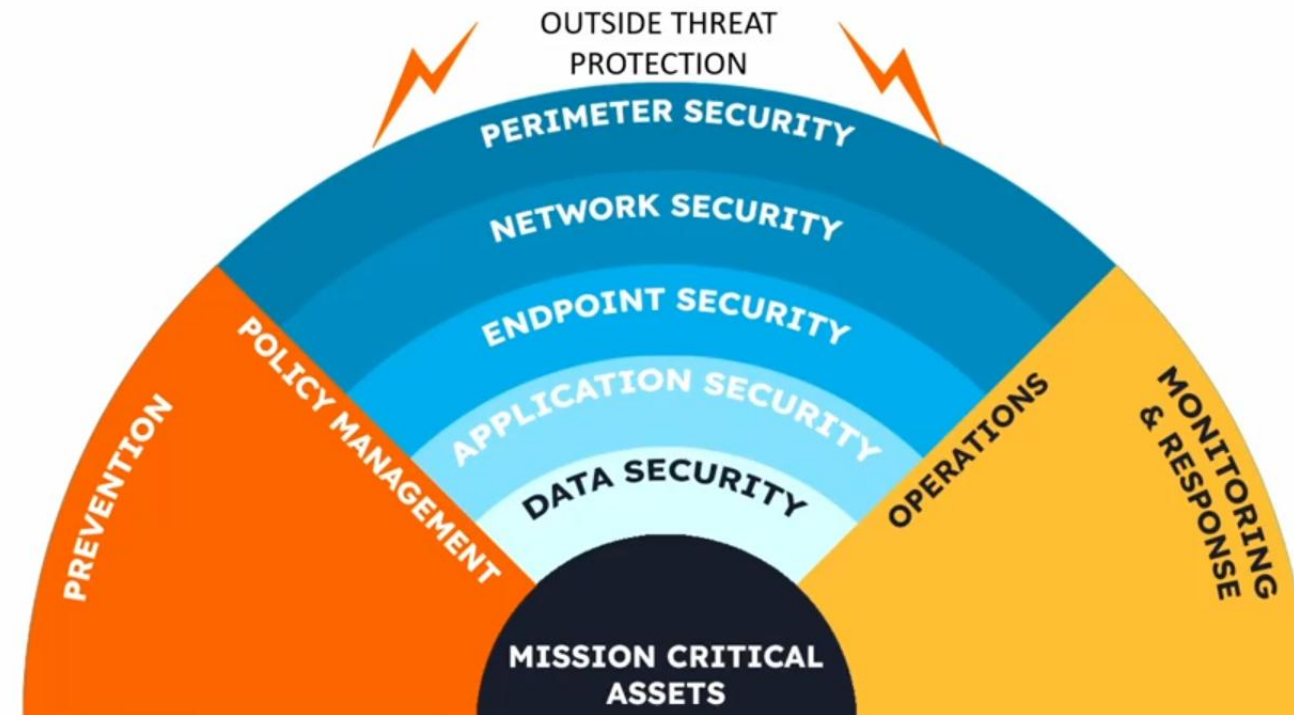
Security Layers (Defense in Depth)

Modern cybersecurity uses multiple layers of protection.

Defense in Depth

- Network security
- Application security
- Endpoint protection
- Identity management

Defense in Depth/ Layered Defense Model



Zero Trust Security Model

In the end of this session, students will be able to:

In the end of this session, students will be able to:

- Explain the modern cybersecurity threat landscape
- Identify common cyberattack techniques
- Describe the Zero Trust security model
- Understand cloud security challenges
- Analyze enterprise cybersecurity strategies

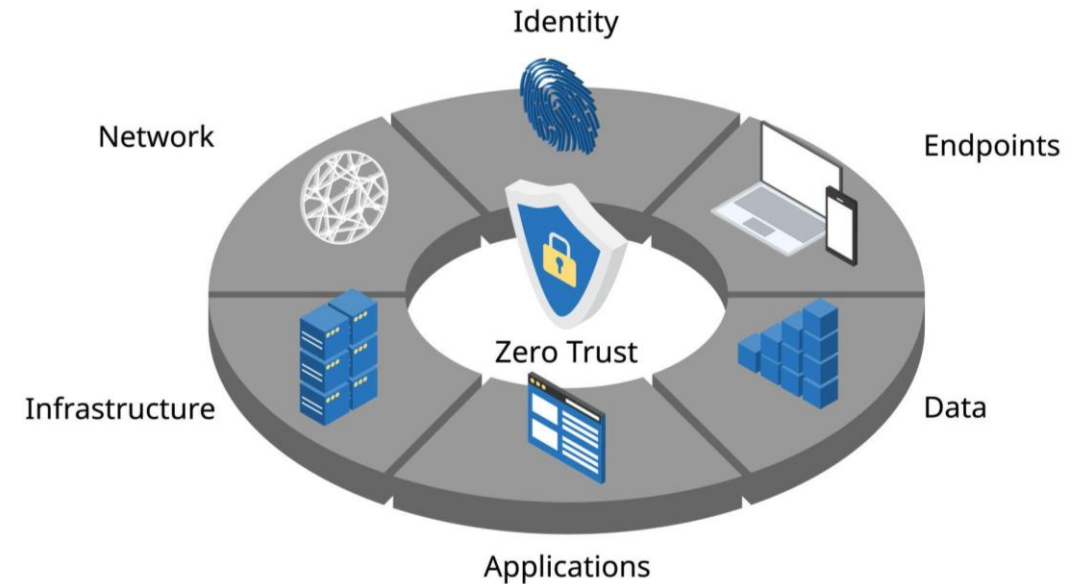


Zero Trust Security Model

Traditional security assumed internal networks were safe. Zero Trust rejects that assumption.

Zero Trust:

- Never trust, always verify
- Every access request must be authenticated
- Continuous monitoring of users and devices



Multi-Factor Authentication (MFA)

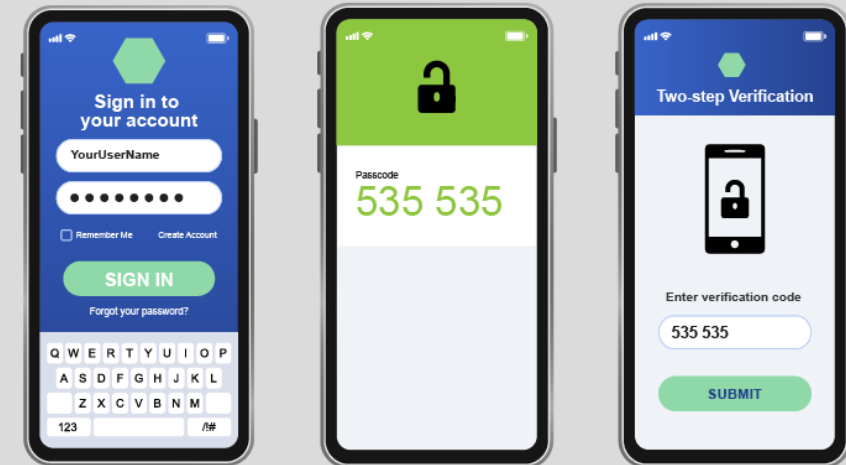
MFA adds additional verification layers to login systems.

Multi-Factor Authentication:

- Something you know (password)
- Something you have (phone token)
- Something you are (biometrics)

Using A Verification Passcode From Your Authenticator App

- Step 1. Enter username and password
Step 2. Get a passcode from your app and enter it



Security Operations Centers

Large organizations use specialized teams to monitor cyber threats.

Security Operations Center (SOC):

- Monitors network activity
- Detects suspicious behavior
- Responds to security incidents

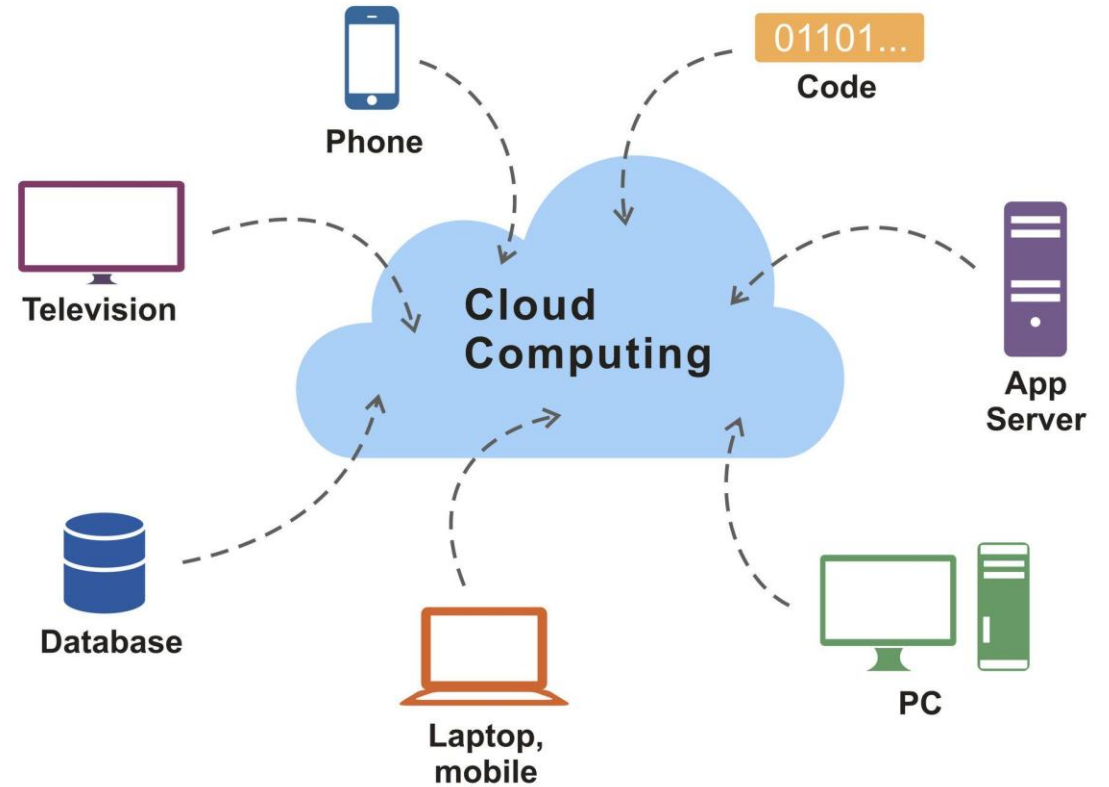


Introduction to Cloud Computing

Cloud computing changed how organizations manage infrastructure.

Cloud Computing:

- On-demand computing resources
- Internet-based infrastructure
- Scalable storage and processing



Cloud Security Challenges

Cloud systems introduce new vulnerabilities.

Cloud Security Issues:

- Misconfigured storage
- Shared infrastructure risks
- Identity management complexity

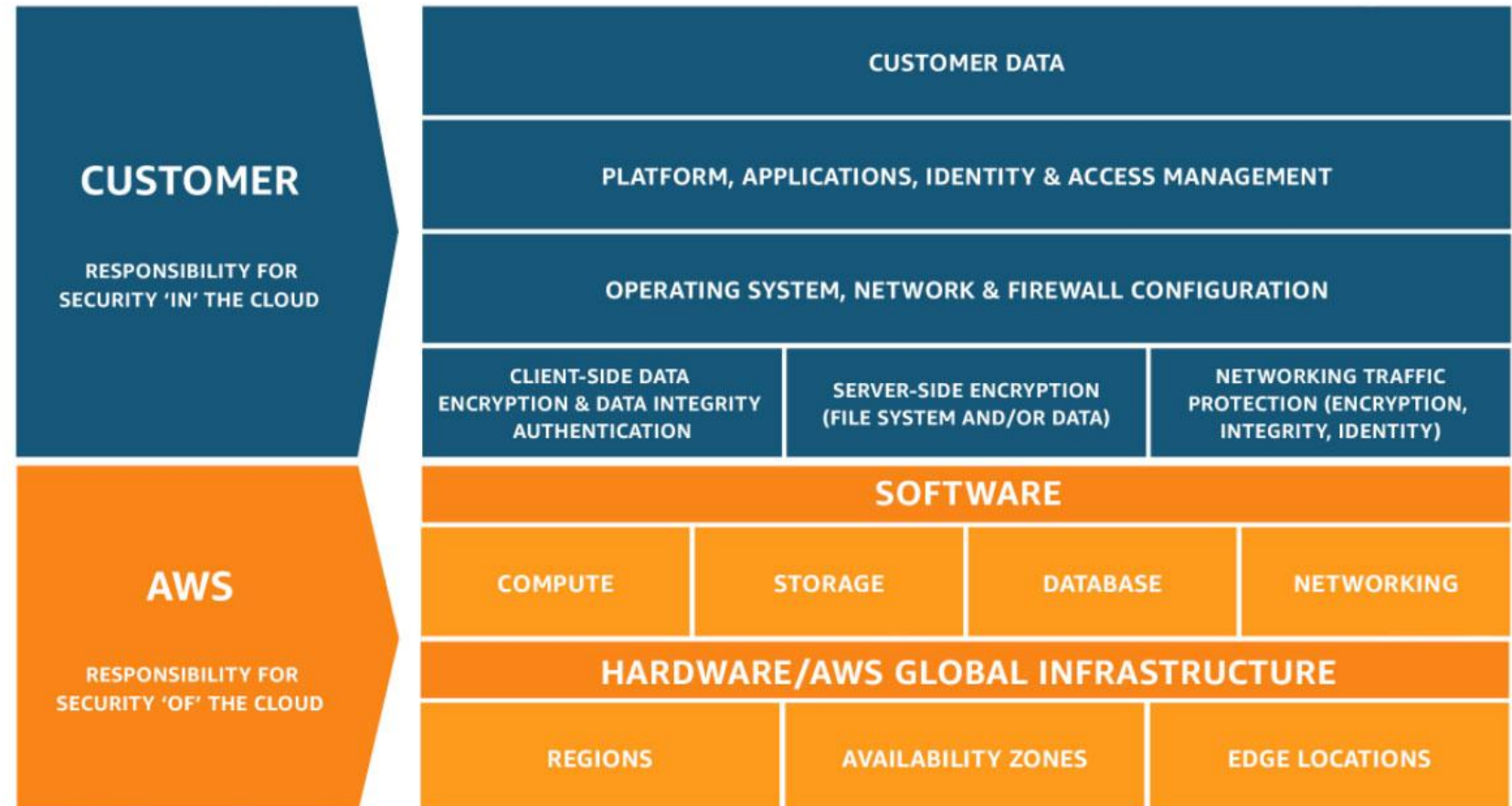


Shared Responsibility Model

Cloud providers and customers share security responsibilities.

Shared Responsibility:

- Provider secures infrastructure
- Customer secures applications and data



How does Docker Work ?

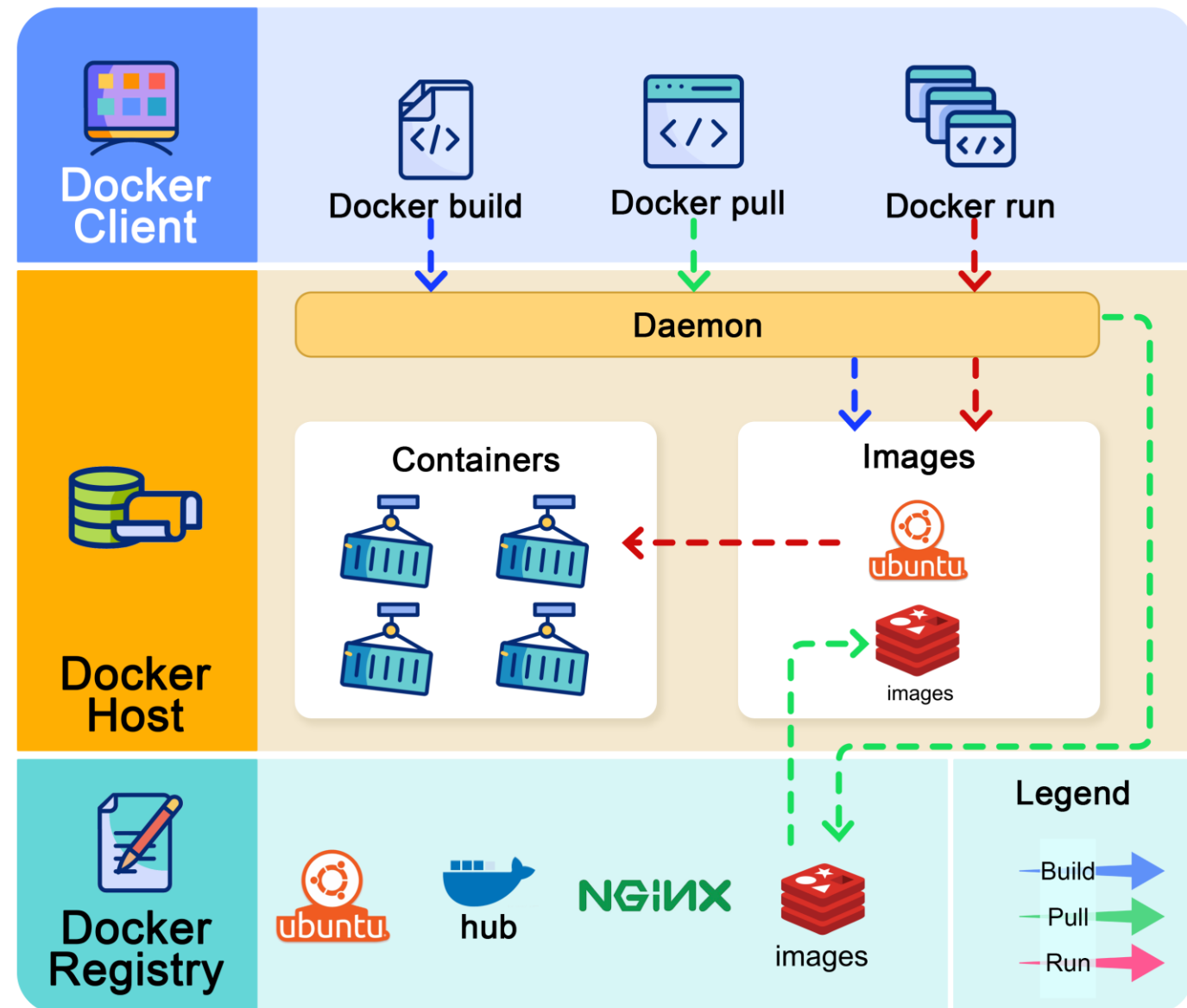
 blog.bytebytego.com

Containerization

Containers help deploy applications efficiently.

Containers:

- Lightweight virtualization
- Isolated application environments
- Common tool: Docker



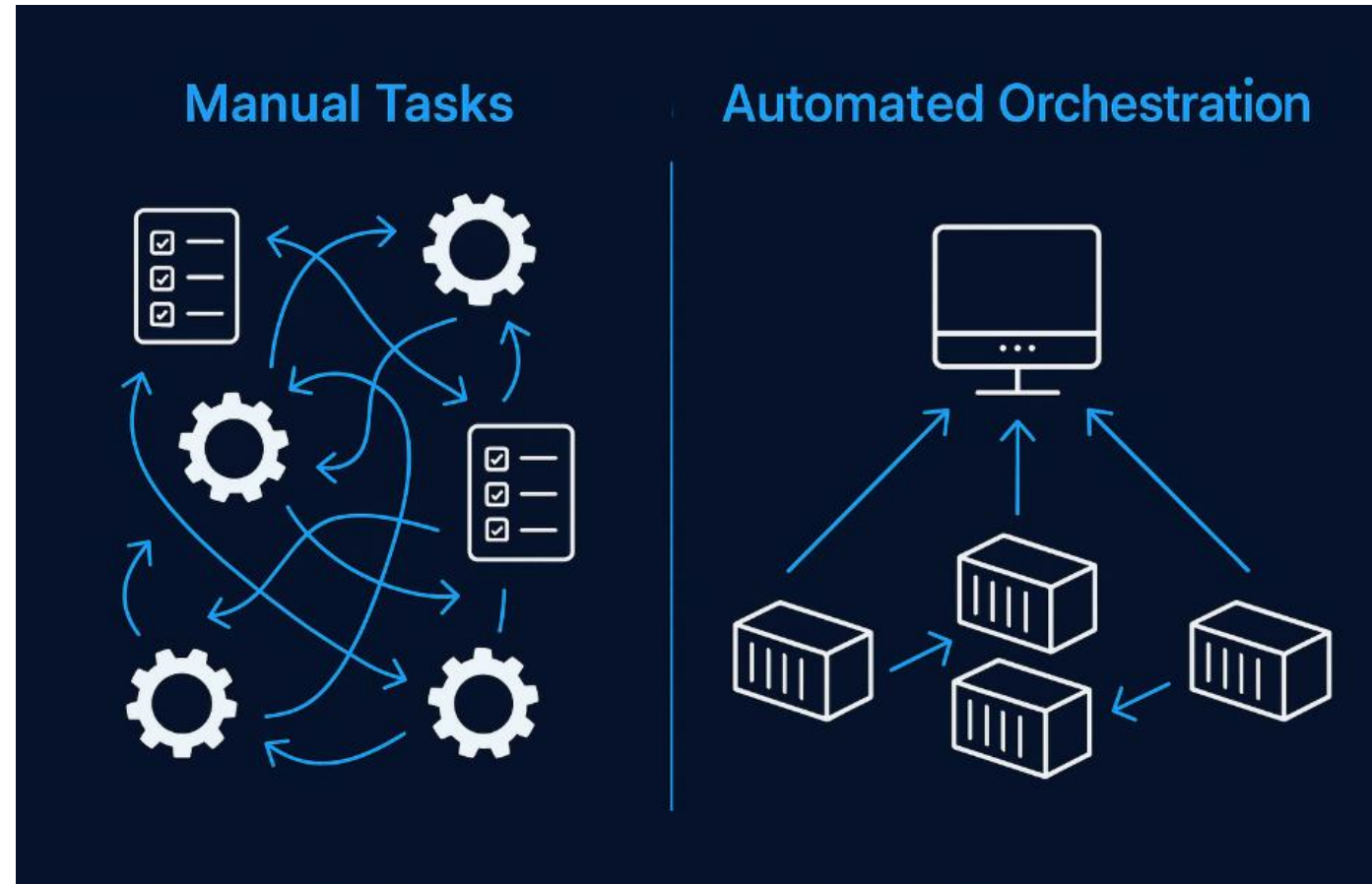
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
UNISAN, QUEZON BRANCH

Kubernetes

Kubernetes manages containerized applications at scale.

Kubernetes:

- Container orchestration platform
- Automates deployment and scaling
- Widely used in cloud infrastructure



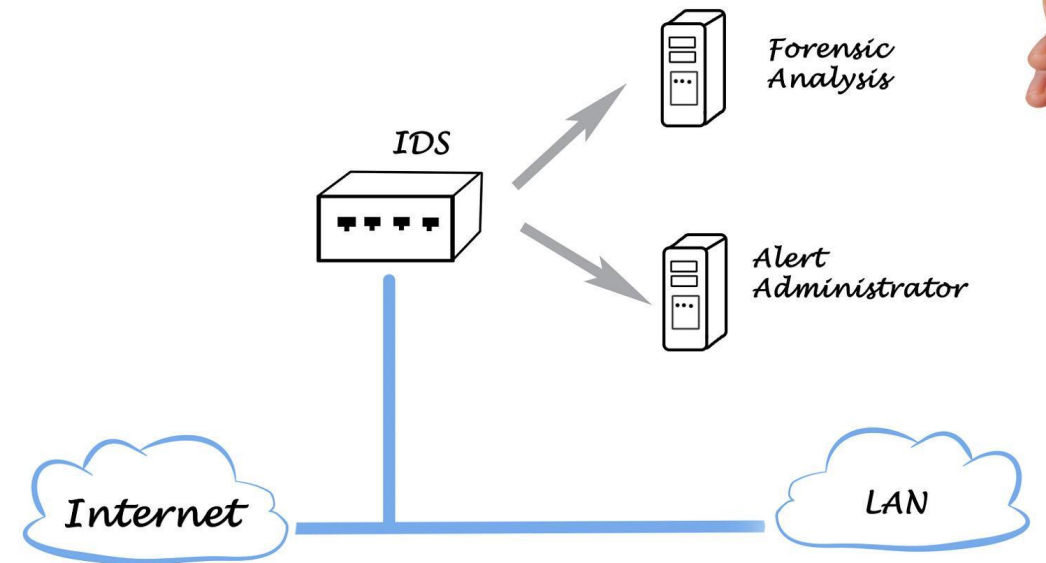
Intrusion Detection Systems (IDS)

Organizations must constantly monitor network traffic to detect potential cyber attacks. Intrusion Detection Systems help identify suspicious activities before they cause damage.

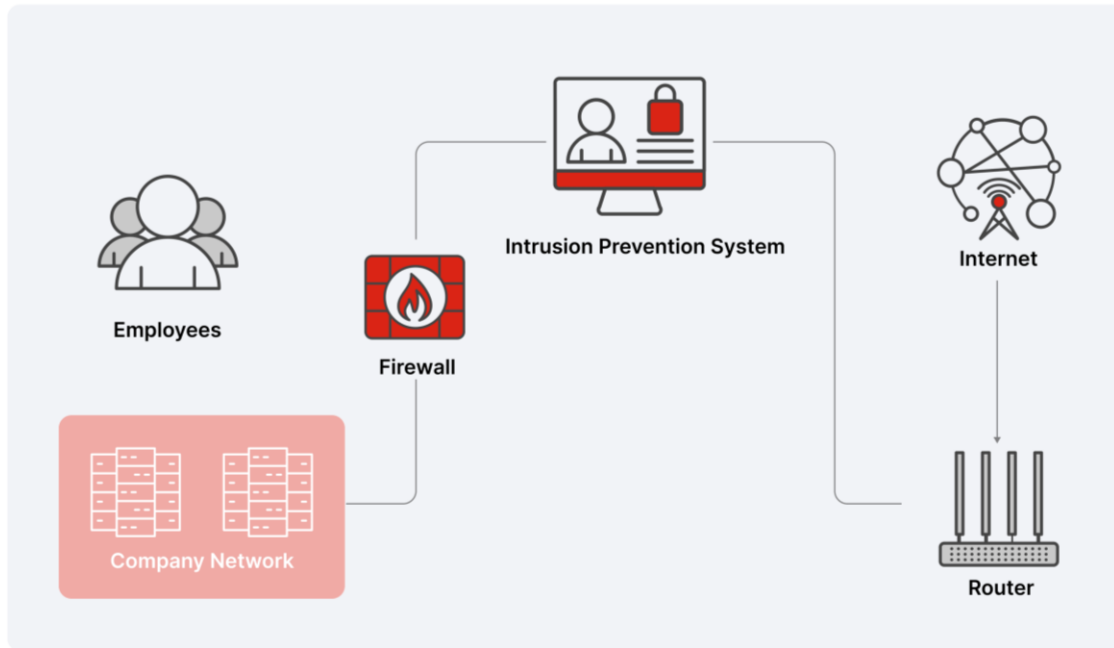
Intrusion Detection System (IDS):

- Monitors network traffic for malicious activity
- Detects unusual patterns or anomalies
- Sends alerts to security teams

Intrusion Detection System



What Is an **Intrusion Prevention System** (IPS)?



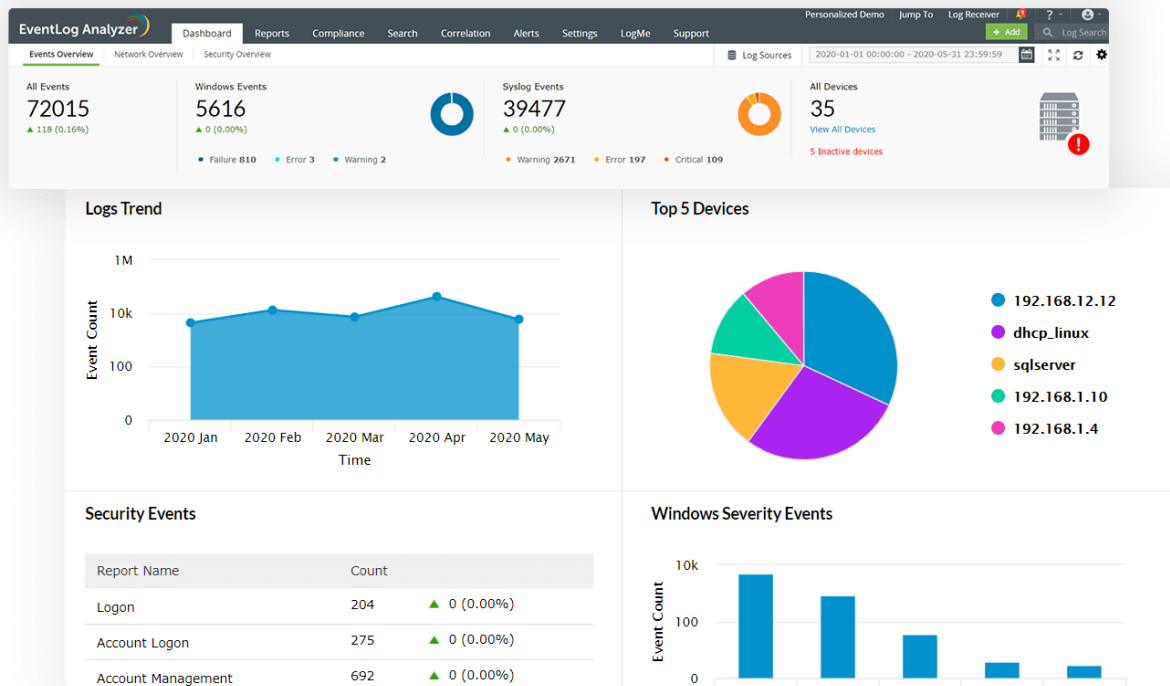
Intrusion Prevention Systems (IPS)

While IDS detects threats, IPS goes further by actively blocking malicious activity.

Intrusion Prevention System:

- Detects and blocks malicious network traffic
- Automatically stops suspicious connections
- Protects systems in real-time





Security Information and Event Management (SIEM)

Large organizations generate massive amounts of security data. SIEM systems analyze logs and security alerts across the entire network.

SIEM:

- Collects logs from multiple systems
- Analyzes security events
- Detects suspicious behavior patterns

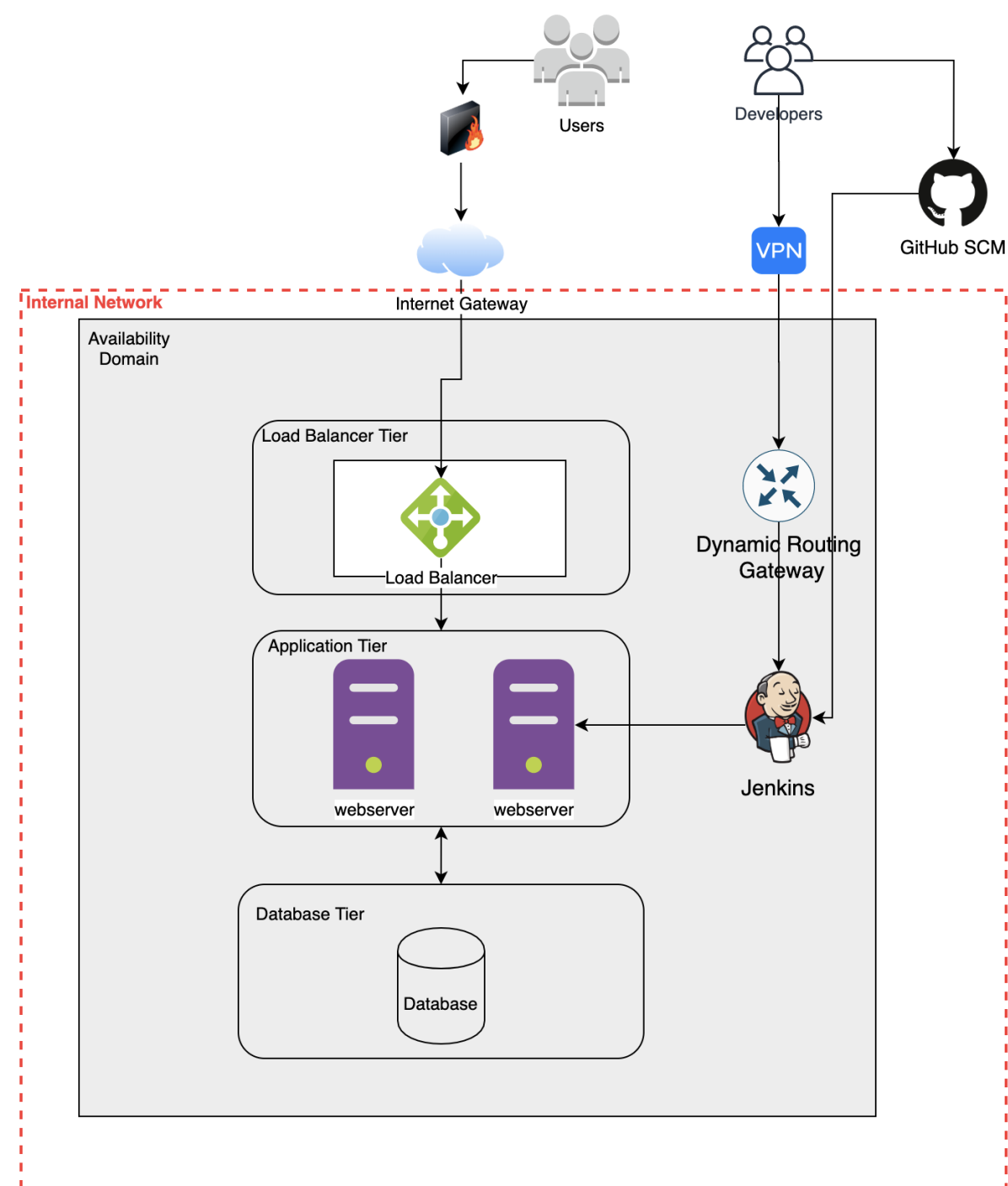


Cloud-Native Security Tools

Cloud environments require specialized security tools designed for distributed systems.

Cloud Security Tools:

- Cloud Security Posture Management (CSPM)
- Container security scanners
- Identity and access monitoring



Ethical Hacking

Organizations hire ethical hackers to test system defenses before attackers do.

Ethical Hacking:

- Authorized penetration testing
- Identifies security weaknesses
- Helps organizations improve defenses



Cybersecurity Careers

Cybersecurity is one of the fastest-growing fields in IT.

Cybersecurity Roles:

- Security Analyst
- Penetration Tester
- Security Architect
- Incident Response Specialist



Future Cybersecurity Trends

Cybersecurity continues evolving as technology advances.

Future Security Trends:

- AI-powered cyber attacks
- AI-driven threat detection
- Zero Trust becoming standard
- Increasing cyber warfare risks



Key Framework for Evaluating Security Strategies

Organizations evaluate cybersecurity strategies through multiple dimensions.

Security Evaluation Framework:

- Risk level of assets
- Threat likelihood
- Security control effectiveness
- Incident response capability



Final Synthesis

Cybersecurity is not just about technology but also strategy, behavior, and governance.

Understanding security issues is essential for modern IT professionals.

Key Takeaways:

- Cyber threats are increasing globally
- Human behavior often causes security breaches
- Zero Trust improves system protection
- Cloud security requires shared responsibility
- Cybersecurity skills are essential for IT careers



Module 3 Activity (Group)

Cyber Incident Case Analysis

Guidelines/Requirements:

1. For a group of 4-5 students (one submission per group only)
2. Write a case analysis covering:
 - Analyze a real cyberattack case in the Philippines
 - Explain how the attack happened
 - Identify security failures
 - Propose prevention strategies
 - Cite at least 3 credible reports

